

Bonus task 4 2.6.2026

Beam structure

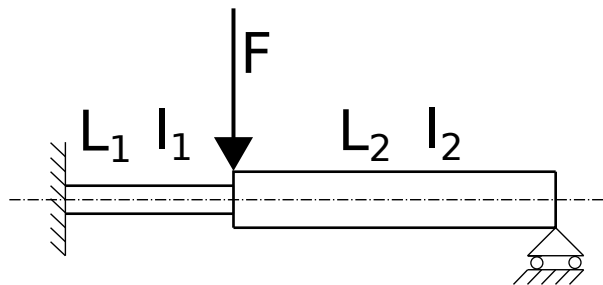
Possibility to earn five bonus points

Deadline: Before the lectures on the 4th of June

Derive the elemental stiffness matrix $\overline{\mathbf{K}}$ for a two-dimensional two-noded Euler-Bernoulli beam element (4 DOFs: $[u_{Y1}, \theta_{Z1}, u_{Y2}, \theta_{Z2}]$) using either MATLAB syms or Python (SymPy).

Then use the derived matrix to assemble a two-element mesh (thus three nodes, two DOFs per node) for the beam, apply the given loading with external force F , impose appropriate boundary conditions, and solve the global nodal responses. Report u_Y and θ_Z at the node where F is applied.

Finally, compute the member (elemental) forces for all elements.



The parameters of the problem are given in Table.

	Young's modulus, E [MPa]	Second moment of area, I [mm ⁴]	Length, L [mm]
Beam 1	210000	4762	1000
Beam 2	210000	9524	2000
Force, F [N]	100		

Write a short report (submission in **PDF format**). Also, submit all necessary MATLAB/Python files and verify that they run without issues. Bonus points are awarded as follows:

- **Displacement field approximation v_h via Eq. (5) (1 point).** Modify and run `DeriveStiffnessMatrixForEB4DOF_template.m` (MATLAB) or `DeriveStiffnessMatrixForEB4DOF_template.py` (Python). Test the function using `run_test_template.m` (MATLAB) or `run_test_template.py` (Python) by imposing a downward displacement of 1 mm at node 2, and evaluate the vertical displacements $v_h(500)$ and $v_h(750)$.
- **Element internal force vector (1 point).** Complete the strain energy formulation and implement the function that returns the element internal force vector (Eq. (10)) in `DeriveStiffnessMatrixForEB4DOF_template.m` or `DeriveStiffnessMatrixForEB4DOF_template.py`. Test the function with the same setup (1 mm downward at node 2) and report the vector of internal forces.
- **Global assembly after elimination of constraints (1 point).** Assemble the global system after applying boundary conditions, compute the determinant of the reduced stiffness matrix, and submit the value.
- **Global nodal displacements at node 2 (1 point).** Report the correct global nodal displacement and rotation at node 2 (Be careful with signs).
- **Member (elemental) forces (1 point).** Report the element (member) force vectors for all elements, and clearly state the signs (- or +) and what does it mean (compression or elongation).

Tips for an elemental stiffness matrix (done partly during lecture)

Two node 4 DOF Euler-Bernoulli type beam element has displacement v_1 and rotation θ_1 at node 1 ($x = 0$) and v_2 and rotation θ_2 at node 2 ($x = L$). So it is assumed that the element's local coordinate system $\{x, y\}$ is at an element's node 1 ($x = 0$).

Four polynomials are needed to approximate the displacement field v_h in the vertical direction over the element's length (material coordinate x).

$$v_h = a_0 + a_1x + a_2x^2 + a_3x^3 \quad (1)$$

Rotation θ can be approximated from the displacement field as follows:

$$\theta \approx \frac{dv_h}{dx} = a_1 + 2a_2x + 3a_3x^2 \quad (2)$$

Then giving boundary conditions at elemental level $v_h(x = 0) := v_1$, $\theta(x = 0) := \theta_1$, $v_h(x = L) := v_2$ and $\theta(x = L) := \theta_2$:

$$\begin{cases} v_h(0) := a_0 = v_1 \\ \theta(0) := a_1 = \theta_1 \\ v_h(L) := a_0 + a_1L + a_2L^2 + a_3L^3 = v_2 \\ \theta(L) := a_1 + 2a_2L + 3a_3L^2 = \theta_2 \end{cases} \quad (3)$$

Eqs. (3) in a matrix form:

$$\underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & L & L^2 & L^3 \\ 0 & 1 & 2L & 3L^2 \end{pmatrix}}_{\mathbf{A}} \underbrace{\begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}}_{\mathbf{a}} = \underbrace{\begin{pmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{pmatrix}}_{\mathbf{u}} \quad (4)$$

then solving $\mathbf{a} = \mathbf{A}^{-1}\mathbf{u}$ and substituting them into Eq.(1).

$$v_h = (\mathbf{A}^{-1}\mathbf{u})^T \mathbf{p} = \underbrace{\mathbf{p}^T \mathbf{A}^{-1}}_{\mathbf{N}} \mathbf{u} \quad (5)$$

where $\mathbf{N}$ includes shape functions (known as ansatz functions in German literature).

Strain energy and elemental stiffness matrix

Due to the kinematic assumptions in Euler-Bernoulli theory, vertical displacement u at distance y from the beam's neutral axis can be written as:

$$u(x) = -y \frac{\partial v}{\partial x} \quad (6)$$

and strain

$$\varepsilon = \frac{\partial u}{\partial x} = -y \frac{\partial^2 v}{\partial x^2} = -y \underbrace{\frac{\partial \theta}{\partial x}}_{\varepsilon_{xx}^1} \quad (7)$$

Using the assumptions, strain energy can be written as:

$$W_{int} = \frac{1}{2} \int_V \sigma \varepsilon dV = \frac{1}{2} E \int_0^L \varepsilon_{xx}^1{}^2 \int_A y^2 dA dx = \frac{1}{2} EI \int_0^L \varepsilon_{xx}^1{}^2 dx \quad (8)$$

where $I = \int_A y^2 dA$ is moment of inertia and ε_{xx}^1 is curvature. Curvature (kinematic measure $\varepsilon_{xx}^1 = 1/\text{radius of a curved beam}$) tells how much the beam has deformed by bending moment and it is approximately equal to the second derivative of the deflection with respect to material coordinate x :

$$\varepsilon_{xx}^1 \approx \frac{d\theta}{dx} = \frac{d^2 v_h}{dx^2} \quad (9)$$

The vector of elemental internal forces:

$$\mathbf{f}_{int} = \frac{\partial W_{int}}{\partial \mathbf{u}} \quad (10)$$

Elemental local stiffness matrix:

$$\bar{\mathbf{K}} = \frac{\partial \mathbf{W}}{\partial \mathbf{u}} \quad (11)$$

Analyzed structures are not often parallel to the element's longitudinal direction -> Transformation from a local elemental coordinate system to a global coordinate system.

PLEASE NOTE: In this task, the element's local coordinate system is parallel to the beam's global coordinate system, so no transformation is required. You could still perform one, but the element-level global stiffness matrix would then contain zero rows and columns that must be eliminated.

Assembling

Sometimes it is helpful to present elements' connectivity and corresponding nodal displacements in a table as follows:

	Node I	Node J	Global nodal displacements	Local nodal displacements
Beam 1	1	2	$u_{X_1}, \theta_{Z_1}, u_{X_2}, \theta_{Z_2}$	$v_1, \theta_1, v_2, \theta_2$
Beam 2	2	3	$u_{X_2}, \theta_{Z_2}, u_{X_3}, \theta_{Z_3}$	$v_3, \theta_3, v_4, \theta_4$

It can be seen from the connectivity table what stiffness components of global elemental stiffness matrices $\mathbf{K}_G^{(1)}$ and $\mathbf{K}_G^{(2)}$ are needed to sum. To make assembling, you need to keep in mind that the global stiffness matrix $\mathbf{K}_G$ for the analysed beam structure should be a size of 6×6 (number of nodes $\times$ 2 DOFs at a node).

Boundary conditions and linear solution

After eliminating rows and columns of the global stiffness matrix and external force vector, global displacements can be solved:

$$\mathbf{u}_{G_{red}} = \mathbf{K}_{G_{red}}^{-1} \mathbf{f}_{G_{red}}$$

A size of $\mathbf{K}$ is 3×3 (all nodal displacements (DOF's) - three constraints = 3).

Member forces

Firstly you need to define global displacements for all elements. For example for element 1:

$$\mathbf{u}_G^{(1)} = \begin{pmatrix} u_{X_1} \\ \theta_{Z_1} \\ u_{X_2} \\ \theta_{Z_2} \end{pmatrix}$$

Then you need to present those in the elemental level. In this case, the element's local coordinate systems are parallel with the global coordinate system so no need for transformation i.e. $\mathbf{u}_G^{(i)} = \bar{\mathbf{u}}^{(i)}$. Member forces (for an element i) can be found as

$$\bar{\mathbf{f}}^{(i)} = \bar{\mathbf{K}}^{(i)} \bar{\mathbf{u}}^{(i)} \quad (12)$$